



# Mr. SOK PHANIT

Power Electronics, Electrical  
Design of PCB, Deep learning (AI)  
Python, C, and MCU programming  
Research & Development (R&D)

## Contact



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## Education

### Mechatronic Engineering

Rajamangala University of Technology  
Thanyaburi, Thailand

Bachelor's degree. 2018 – 2022

GPA 3.88/4.0 with First-Class Honors

### Mechatronic Engineering

Gyeongsang National University, South Korea

Master's degree. 2023 – 2025

GPA 4.43/4.5 with IEEE SCI(E) journal

## Award

- 31<sup>st</sup> Young Pioneer Researcher Award
- Best Conference Paper Award
- Award for Excellent Academic Achievement
- Grade A of diploma/Cambodia
- Top 7<sup>th</sup> of Outstanding physics students

## Skills

- PCB design using Altium Designer
- Power converter system design
- PSIM, TINA-TI, PSpice, LTspice Simulation
- and analysis of circuits
- Reliability analysis of an electronic circuit
- MCU programming & sensor data analysis
- Python, C programming
- Python for deep learning (AI)
- Electrical panel wiring (HV/LV/PLC systems)
- 3D design in SolidWorks
- Various experimental equipment
- Computer skills and Microsoft Word

## Language

- Khmer – Native language
- Thai – Fluent (all aspects)
- English – Fluent (all aspects)
- Lao – Fluent in Communication
- Korean – (TOPIK II, Level 3)

## About Me

A Mechatronics Engineer from Cambodia with a strong academic foundation and a deep passion for technological innovation. Completed master's and bachelor's degrees in South Korea and Thailand. I have published multiple conference papers, IEEE SCI journals, and patents. I specialize in power electronics, electrical panel wiring, protection circuit, and AI-based fault diagnosis. I have practical experience in PCB design, circuit simulation, and microcontroller programming, with strong skills in Python and C, especially in deep learning, and proficient in general computer skill use and Microsoft Office. Driven to create reliable, efficient solutions that integrate academic research with practical industrial needs. Committed to advancing technology and sustainability while continuously evolving as a global researcher and engineer through innovation, collaboration, and resilience.

## Professional Experience

**Survitech Co., Ltd, Thailand, July 2021 – November 2021**

Junior PCB design, Internship program

- Design a PCB prototype using Altium Designer.
- Write code for a touchscreen HMI user interface.
- Program microcontrollers both in Arduino IDE and Micro Python.

**Power Electronics and Reliability Lab, GNU, August 2023 – August 2025**

Full-time researcher

- Designed a voltage clamping circuit for DC solid-state breakers.
- Developed and performed reliability analysis of a converter/hybrid converter/PFC Converter.
- AI-based for life prediction using the LSTM algorithm in AAV.
- DSP and various microcontroller coding.
- AI Research of optimized image classification FPS on Jetson Nano.

**R&D H/W Engineer, Hyowon PowerTech, October 2025 – Presents**

Hardware Design, Testing & Performance Evaluation

- HV EV Inverter Circuit/PCB Design Assistance
- PCB & Component Review / Modification
- EV Inverter Hardware Performance Testing
- Protection Function Design/Testing (OCP, OVP, DESAT, SC)
- H/W Fault Diagnosis/Report trigger protection
- The power loss analysis of EV Inverter using DPT

## References

### Kang Feel-Soon

Professors/Gyeongsang National University

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## Analysis and Implementation Clamping Circuit for Surge Vo in the DC Solid-State Circuit B

PHANIT SOK<sup>1</sup>, SUNG-GEUN SONG<sup>2</sup>, (Member, IEEE), AND FEEL-SOON KANG<sup>3</sup>, (Member, IEEE)<sup>1</sup>Department of Mechatronics Engineering, Gyeongsang National University, Jinju-si 52725, So

Young Gyeongsang Research Center, Korea Electronics Technology Institute, Gwangju 61011,

Corresponding author: Feel-Soon Kang (feelsoon@gnu.ac.kr)

This work was supported by the Technology Innovation Program (Development of Technology) funded by the Ministry of Trade, Industry and Energy (MOTIE), Sou

**ABSTRACT** The DC Solid-State Circuit Breaker (SSCB) is equipment and ensure system stability by rapidly interrupting the surge voltage, driven by high switching speeds and the stray can easily damage semiconductor switches in DC SSCB. Due to effectively suppressing surge voltage and addressing this problem analyzed and simulated in this paper involves using a crowbar protection into three MOVs. This approach employs lower 1 voltage achieved by double MOVs in series with a thyristor, a protection. It is facilitated by a voltage divider circuit composed is also used for energy absorption from the stray inductance computer-aided simulations and small-scale experiments. A simulation of the MOV used in this paper for simulation is determined algorithm for optimization.

**INDEX TERMS** Crowbar clamping circuit, curve fit function suppression, snubber circuit, solid-state circuit breaker (SSCB)

## I. INTRODUCTION

DC power systems are becoming increasingly popular due to their efficiency in distributed power generation solutions suitable for industrial, commercial, and residential sectors, offering enhanced energy efficiency, improved reliability, and greater power management flexibility [1], [2], [3]. These systems are particularly beneficial in electric transportation, including marine vessels, and electric aircraft [4]. However, the adoption of DC power faces significant technical challenges in fault detection and isolation [5]. A traditional

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## Improving Surge Voltage the MOV-MOV-C Approach Solid-State Circuit Bre

PHANIT SOK<sup>1</sup>, SUNG-GEUN SONG<sup>2</sup> (Me AND FEEL-SOON KANG<sup>3</sup><sup>1</sup>Department of Mechatronics Engineering, Gyeongsang Na<sup>2</sup>Energy Conversion Research Center, Korea Electronics Tech<sup>3</sup>Shandong Provincial Key Laboratory of Automotive Electronics Technology, Qilu Un

CORRESPONDING AUTHOR: FEEL-SOON KAN

This work was supported by the Technology Innovation Program under Grant RS-202 Distribution Technology, funded by the Ministry of Trade,

**ABSTRACT** The dc solid-state circuit breaker (SSCB) is and enhance system stability by quickly interrupting fault double metal oxide varistors (MOVs) and the single capacitor by MOV-MOV-C that have been successfully demonstrated effectively suppressing high-frequency oscillations caused low-frequency harmonics introduced by the RC snubber. For improvement, particularly in reducing surge voltage across in on-state normal operation, and discussing the overall article analyzes and compares power loss between different and minimize it. The surge voltage also improves from enhancement incorporating a new MOV featuring a lower at optimizing surge voltage protection, represented by a 1 critical components was also developed to discuss the total of both snubber circuits and validated through the graph. voltage improvements was conducted through small-scale experiments.

**INDEX TERMS** Cost model analysis (CMA), dc circuit overvoltage suppression, power loss, snubber circuit, solid-

## I. INTRODUCTION

The dc grids are widely utilized across various applications, including renewable energy sources, electric vessels, and aircraft propulsion systems [1], [2]. With the growing adoption of dc grids, fault current detection and clearance have become critical focus areas because some units may have more current than devices designed for during a short circuit, leading to potential instability and system damage [3]. The dc circuit breaker (DCCB), a crucial technology for safeguarding dc systems within multiterminal dc networks, must meet several essential requirements [4], [5]. Nevertheless, the absence

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## Reliability Analysis of PFC Converter in LED TVs using Markov Approaches

Phanit Sok, Sung-Yong Joo, and Feel-Soon Kang, Member, IEEE

**Abstract**—The Power Factor Correction (PFC) converter is a crucial part of TVs, and it is responsible for supplying power to both the Light-Emitting Diodes (LEDs) backlight and the motherboard. While performing AC-DC conversion, the PFC converter also provides a feedback signal to the controller to maintain a high PF. A fault in the PFC feedback part can lead to more stress and a higher failure rate in TVs. Thus, this paper presents a reliability analysis of this converter by comparing scenarios with the PFC feedback enabled and disabled. An actual 175 W application board used in the TVs is selected as the experimental platform, where specific points in the circuit are intentionally disconnected to analyze the effects of PFC feedback failure. The reliability analysis calculates the total failure rate and Mean Time to Failure (MTTF) using Markov approaches, with each electronic component's failure rate referring to the FIDES methodology fault library. Additionally, the junction temperatures of MOSFETs and diodes are determined through a thermal model to compare which device heats the most. Finally, the validation of the system with and without the PFC part is conducted through experiments.

**Index Terms**—AC-DC converter, FIDES methodology, LEDs driver, Markov model, Mean Time to Failure (MTTF), Power Factor Correction (PFC), Reliability analysis, TVs.

## I. INTRODUCTION

Lighting plays a vital role in daily life, with electric lighting consuming about 20% of global electricity, and Light Emitting Diodes (LEDs) have emerged as a preferred solution due to their long lifespan and high efficiency [1]. The efficiency of lighting systems depends not only on the LED types but also heavily on the driver circuit. Since LEDs operate best with dc voltage, an ac-dc conversion stage is required to convert the input ac while maintaining a high unity power factor (PF). Among various solutions, switching converters are widely used for their high efficiency in driving LEDs [2].

A prior study proposed a single-stage PFC LED driver using a Cuk converter with hysteresis control, achieving fast response and high efficiency [3]. An evolved design synthesized a buck-boost and Class-E resonant converter using a single switch, achieving soft-switching, 0.995 PF, and 90.8% efficiency [4].

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A further single-stage LED driver for street lighting achieves 91.1% efficiency by integrating dual boost circuits with a shared inductor and half-bridge LLC converter switches. [5]. A two-stage automotive LED driver using a noninverting buck-boost front end and multiple LCL-T resonant converters achieves 93% efficiency [6]. An additional topology combined a CLLC resonant dc-dc converter with a PFC stage, optimizing performance and reaching roughly 95% efficiency [7].

To enhance efficiency, most LED drivers are designed with PFC feedback by correcting the shape of the current drawn from the ac mains, making it more sinusoidal and in phase with ac [8]. However, if the feedback signal fails, the overall system failure rate can significantly increase and may lead to component overheating. Therefore, these reasons motivate the reliability analysis of the LED driver used in TVs since it is an LED backlight is one of the specific applications. The reliability analysis of one converter method used in a prior study includes Part Count Analysis (PCA), Part Stress Analysis (PSA), and Fault Tree Analysis (FTA) [9]. PCA offers quick early-stage estimates, PSA provides realistic component reliability under stress, and FTA analyzes functional system-level failures for complex designs under logic gate connections [9]. The advanced technique used for reliability analysis is the Markov model, which predicts system reliability by modeling transitions between fault states over time based solely on the current state [10]–[12].

This paper analyzes the reliability of the PFC converter in TVs using a Markov model. It compares the total failure rate and Mean Time to Failure (MTTF) between PFC feedback function enabled and disabled by modifying a connection point on the actual board. Additionally, this paper estimates the junction temperatures of the MOSFETs and diodes.

## A. Contribution

The previous research used PSA technical methods in the reliability analysis of the PFC multilevel rectifier-based LED driver, as presented in [13]. The novelty of this paper involves using the Markov model to analyze reliability, focusing on the PFC converter configured as a hybrid three-stage combined converter multi-channel output LEDs backlight, and power delivery to the motherboard in TVs.

Phanit Sok and Feel-Soon Kang are with the Department of Mechatronics Engineering, Gyeongsang National University, Jinju, 52725, South Korea (sokphant@gnu.ac.kr, feelsoon@gnu.ac.kr). Sung-Yong Joo is with the Department of Power Electronics Engineering, Samsung Electronics Co., Ltd., Suwon, 16677, South Korea (sungyong.joo@samsung.com).

[1] **P. Sok**, S. -G. Song, and F. -S. Kang, "Analysis and Implementation of the Crowbar Clamping Circuit for Surge Voltage Reduction in the DC Solid-State Circuit Breaker," IEEE Access, vol. 12, pp. 149018-149031, 2024. **SCI(E)**

- This paper proposes a surge voltage suppression method using a crowbar clamping circuit with three MOVs, two lower-rated MOVs in series with a thyristor handle clamping, assisted by a resistor divider. A higher-rated MOV absorbs energy from stray inductance.

[2] **P. Sok**, S. -G. Song, G. Zhou, and F. -S. Kang, "Improving Surge Voltage and Power Loss of the MOV-MOV-C Approach for Low-Scale DC Solid-State Circuit Breaker Applications," IEEE Open Journal of the Industrial Electronics Society, vol. 6, pp. 708-721, 2025. **SCI(E)**

- This article compares power losses across semiconductor switches and proposes a Triple-MOV-C snubber with a lower-rated MOV to enhance surge voltage protection from the conventional MOV-MOC-C approach

[3] **P. Sok**, S. -Y. Joo and F. -S. Kang, "Markov Model-Based Reliability Analysis of LED TVs on PFC Converters With Control Loop Effects," IEEE Transactions on Consumer Electronics, vol. 72, no. 1, pp. 51-64, Feb. 2026. **SCI(E)**

- This paper analyzes an actual board to determine how faults in the PFC feedback signal section affect the converter's failure rate and lifespan, and to propose enhancing reliability in real-world TV applications.

# CONFERENCES PAPERS

## The crowbar clamping circuit to reduce the surge voltage in the DC solid-state circuit breaker

Phanit Sok<sup>1</sup>, Sun-Pil Kim<sup>2</sup>, Sung-Geun Song<sup>3</sup> and Feel-soon Kang<sup>1</sup>

<sup>1</sup>Dept. of Mechatronics Engineering, Gyeongsang

<sup>2</sup>Dept. of Electrical Engineering, Mokpo N

<sup>3</sup>Energy Conversion Research Center, Korea Electron

### Abstract

The surge voltage caused by the metal oxide varistor (MOV) characteristics easily damages semiconductor switches in the DC solid-state circuit breaker (SSCB). This paper presents a surge voltage suppression method for DC SSCB using a crowbar clamping circuit by splitting the overvoltage protection into three separate varistors. The lower DC-rated voltage of double varistors serially optimizes for the surge voltage protection allowed by the voltage divider circuit of two resistors. After theoretical analysis, it is verified by computer-aided simulations.

**Key words:** Metal Oxide Varistor (MOV), Solid-State Circuit Breaker (SSCB), PSpice, Surge voltage

### Introduction

DC systems are helpful in electric transportation applications such as marine and electric aircraft. The demand is continuously increasing, especially in DC solid-state circuit breakers (SSCBs). The research of DC SSCB is designed to detect and isolate faults and safeguard loads and system equipment from excessive current damage [1]. The DC SSCB can detect fault current in less than microseconds but faces surge voltage issues [2]. In [1], it has researched voltage overshoot suppression using RC snubber circuits. The other method optimizes voltage by dividing overvoltage protection and energy absorption into two separate parallel varistors, resulting in a lower DC-rated voltage for overvoltage protection and a higher for energy absorption [3]. The disadvantage of using only one varistor for optimizing overvoltage is a small value decrease in surge voltage.

This paper proposes that a crowbar clamping circuit reduces the surge voltage, using three separate varistors to optimize for surge voltage. The lower DC-rated voltage of double varistors is connected to a serial with thyristor to maximize surge voltage protection by depending on the voltage divider circuit of two resistors, the higher DC-rated voltage varistors for energy absorption. The clamping part comprises double varistors, and the crowbar is defined by a voltage divider circuit [4]. The model of the varistor voltage as a constant provides an approximation that was designed [2]. This paper also proposes modeling a varistor as a nonlinear curve V-I different from PSpice for simulation and finding parameters.

### SSCB with proposed crowbar clamping snubber circuit

#### A. How to create a MOV model

The characteristic of the varistor is highly non-linear, and

## FTA기법을 이용한 하이브리드 전력변환시스템 신뢰성 분석

속 파닛<sup>1</sup>, 김선필<sup>2</sup>, 강필순<sup>1</sup>, 박성준<sup>3</sup>  
경상국립대학교<sup>1</sup>, 국립목포대학교<sup>2</sup>, 전남대학교<sup>3</sup>

## Reliability analysis of hybrid power conversion systems by a fault tree analysis approach

Phanit Sok<sup>1</sup>, Sun-Pil Kim<sup>2</sup>, Feel-soon Kang<sup>1</sup>, Sung-Jun Park<sup>3</sup>  
Gyeongsang National University<sup>1</sup>, Mokpo National University<sup>2</sup>, Chonnam National University<sup>3</sup>

### ABSTRACT

This paper performs reliability and MTTF (MTTF) analysis for hybrid Power Conversion Systems (PCS). Those systems include by Maximum Power Point Tracking (MPPT) implemented in a Photovoltaic (PV) DC-DC boost converter, an energy storage capacitor, an isolated DC-DC converter, the PEMFC stack as a power source, and an isolated DC-DC converter. All the components are connected in parallel to a single-phase inverter. To provide using the Fault Tree Analysis (FTA) worst-case operating conditions and high impact of reliability on the entire system component in the power electronic circuit verified on a small-scale graph.

### 1. Introduction

The residential sector accounts for a significant portion of the global energy demand. In the context of technological advancement, the concept of Energy Management System (EMS) has emerged as a solution to the growing concerns over energy management technologies for residential buildings. Because energy consumption in residential buildings is about 22% of the grid increasing [1]. The main point of this system is the hybrid Power Conversion System (PCS) amalgamation of various energy sources designed to optimize energy usage within the residential sector.

In this paper, we will propose a reliability analysis for hybrid PCS by separating three parallel systems and different power sources. The three systems are parallel to a single-phase energy management system between the storage in the residential. For the reliability evaluation by using the Fault Tree Analysis (FTA), the failure rate considering the operation and quantitatively present the Mean Time

전력전자학회 하계학술대회 논문집 2024. 7. 1~4

## 2025 Spring Conference of the Electrical Equipment and Energy Conversion System Division (2025.5.1~3)

## Enhancing the MOV-MOV-C Approach to Reduce Power Loss and Surge Voltage in Low-Scale DC Solid-State Circuit Breakers

PHANIT SOK, SUNG-GEUN SONG\*, FEEL-SOON KANG

Gyeongsang National University

Korea Electronics Technology Institute\*

[1] **P. Sok**, S. -P. Kim, S. -G. Song, and F. -s. Kang, "The Crowbar Clamping Circuit to Reduce the Surge Voltage in the DC Solid-State Circuit Breaker," 10th International Conference on Applied System Innovation (ICASI), Kyoto, Japan, 2024, pp. 297-299.

- This paper presents a surge voltage suppression method using a crowbar clamping circuit composed of three MOVs. Two lower-rated MOVs connected in series with a thyristor provide clamping, supported by a resistor divider, while a higher-rated MOV absorbs energy from stray inductance.

[2] **P. Sok**, S. -P. Kim, F. -S. Kang, and S. -J. Park, "Reliability analysis of hybrid power conversion systems by a fault tree analysis approach," Power Electronics Society Conference Proceedings (KIPE), Jeju-do, South Korea, 2024, pp. 663-664.

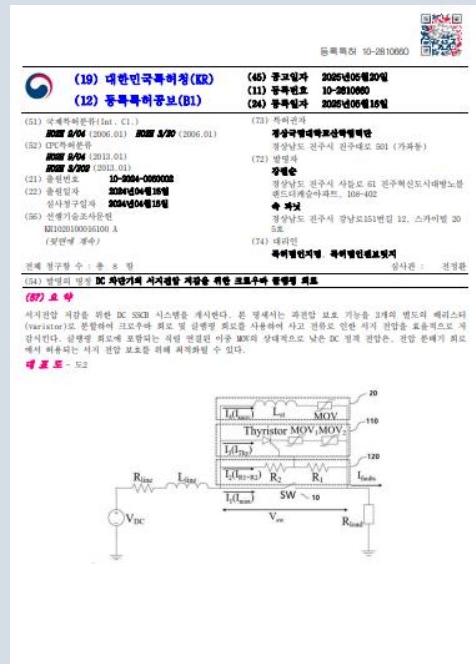
- Designed a 5 kW, 220 VAC inverter for household applications that integrates solar (PV), energy storage (ESS), and fuel cells (PEMFC) to supplement grid power and analyze the reliability of the converter.

[3] **P. Sok**, S. -G. Song, and F. -S. Kang, "Enhancing the MOV-MOV-C Approach to Reduce Power Loss and Surge Voltage in Low-Scale DC Solid-State Circuit Breakers," Spring Conference of the Electrical Equipment and Energy Conversion System Division (KIEE), Gangwon-do, South Korea, 2025, pp. 163-166.

- This article compares power losses in various semiconductor switches and introduces a Triple-MOV-C snubber, featuring a lower-rated MOV, as an improved alternative to the traditional MOV-MOV-C approach for surge voltage protection.

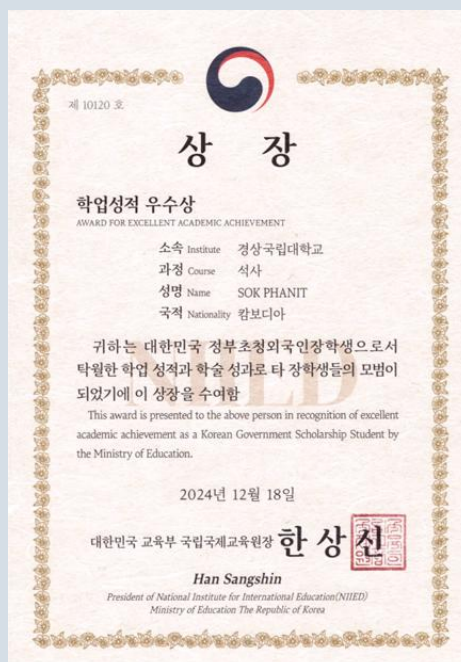


# PATENT



- [1] **P. Sok**, and F. -S. Kang "The crowbar clamping circuit to reduce the surge voltage in the DC solid-state circuit breaker" Patent No. KR102810660B1, May 2025. **[Published]**
- [2] **P. Sok**, and F. -S. Kang "Triple-MOV-C snubber circuit for reducing surge voltage of DC circuit breaker" Patent 2025-1-101-KR, 2025. **[Pending]**

# Award



Honored by Prime Minister Hun Sen



# Professional Experience

## Junior PCB design, Internship program



- Developed PCB prototypes using Altium Designer
- Programmed touchscreen HMI interfaces and installed CNC plasma
- Coded microcontrollers in both Arduino IDE and MicroPython

## Full-time researcher, Power Electronics Lab, GNU



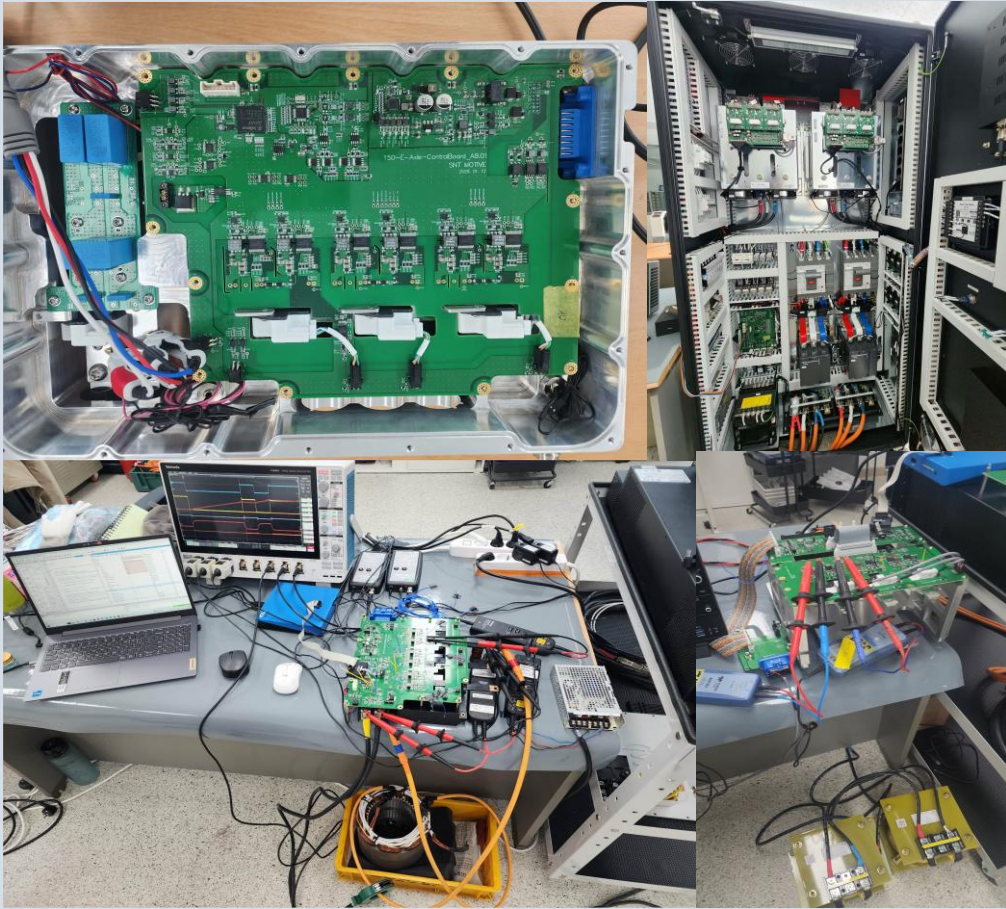
- Designed a voltage clamping circuit for DC solid-state circuit breakers
- Conducted reliability analysis of a hybrid power converter for home energy management
- Performed reliability evaluations across various converter systems
- Developed an AI-based lifespan prediction model using LSTM for AAV applications
- Analyzed the reliability of PFC converters in 4K TVs using a Markov model
- Programmed digital signal processors (DSP) and various microcontrollers



# Professional Experience

## Hardware Design, Testing & Performance Evaluation

### R&D H/W Engineer, Hyowon PowerTech



- HV EV Inverter Circuit/PCB Design Assistance
- PCB & Component Review / Modification
- EV Inverter Hardware Performance Testing
- Protection Function Design/Testing (OCP, OVP, DESAT, SC)
- H/W Fault Diagnosis/Report trigger protection
- The power loss analysis of EV Inverter using DPT

# Other Experiences

세 DIU 2023 - 1798호

한국어강좌 성적증명서  
(Certificate of Transcripts)

성명 : (Name)  문서확인번호 : 2332-9984-3683-2806

생년월일 : (Date of Birth)

국적 : (Nationality)

한국어능력시험 성적증명서  
OFFICIAL TOPIK SCORE REPORT

수험자 정보 (Test-taker's Information)

| 성명            | 성별       | 종시국가                            | 생년월일                     |
|---------------|----------|---------------------------------|--------------------------|
| SOK PHANIT    | 남자(male) | 대한민국<br>KOREA                   | 1999/08/10<br>yyyy/mm/dd |
| 수험번호          | 시험종류     | 시험/시험일                          | 성적유효기간                   |
| 0010328000158 | TOPIK II | 100/2025/05/11<br>th/yyyy/mm/dd | 2027/06/25<br>yyyy/mm/dd |

시험 결과 (Test Result)

| 영역 | Section   | 점수       | 수험자점수 및 수험자평균<br>your score average | 총점        | 등급 |
|----|-----------|----------|-------------------------------------|-----------|----|
| 듣기 | Listening | 48 / 100 | 48.00                               | 130 / 300 | 3급 |
| 쓰기 | Writing   | 32 / 100 | 32.00                               |           |    |
| 읽기 | Reading   | 50 / 100 | 50.00                               |           |    |

언어 수행 능력 (Level Description)

| 시험종류 | Test Type | TOPIK II | 등급 |
|------|-----------|----------|----|
| 듣기   | Listening |          | 3급 |
| 쓰기   | Writing   |          |    |
| 읽기   | Reading   |          |    |

대 전 대 학

국립국제교육원장  
President of National Institute for International Education

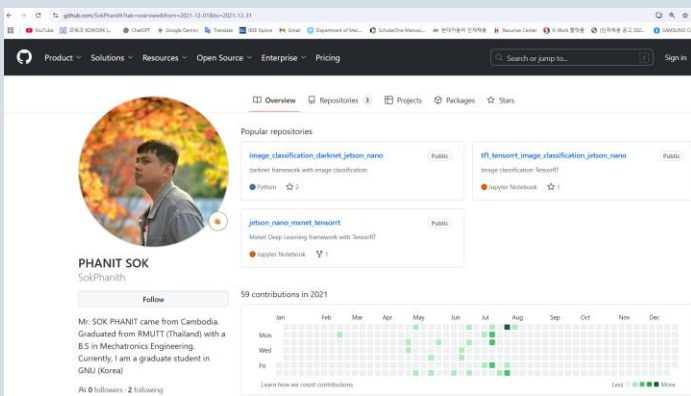
본

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문서 우측 상단의 문서확인번호로 조회 가능합니다.

1 / 2

Daejeon University - Completed Korean language training from September 2022 to August 2023.

Korean language skill: TOPIK II - Level 3.



AI Research of optimized image classification FPS on Jetson Nano.  
[<https://github.com/SokPhanith>]



Volunteer - Cambodian Student Association in South Korea.  
Performance team.